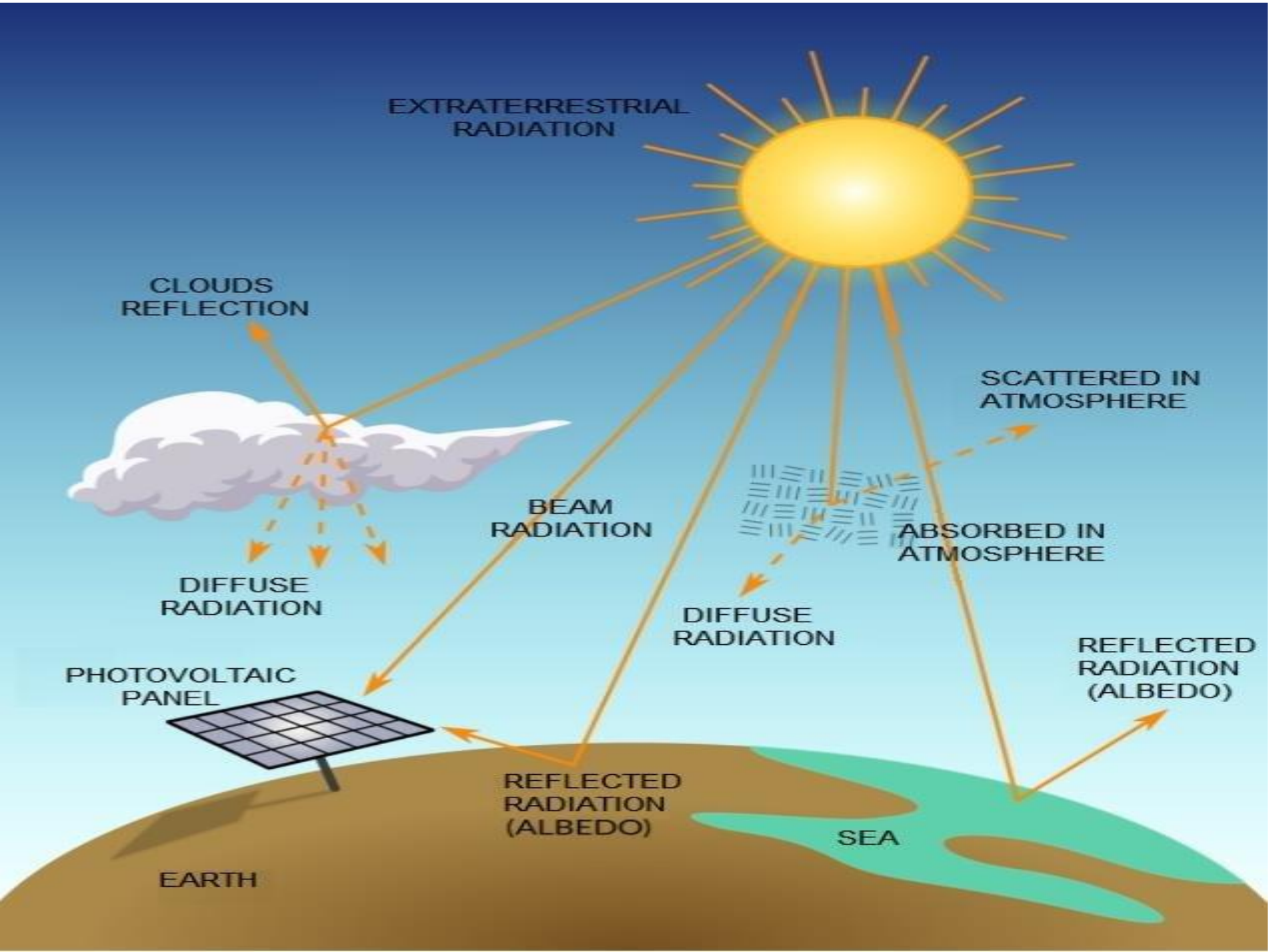
Solar measuring Instruments


EXTRATERRESTRIAL RADIATION
CLOUDS REFLECTION
SCATTERED IN ATMOSPHERE
BEAM RADIATION
ABSORBED IN ATMOSPHERE
DIFFUSE RADIATION
DIFFUSE RADIATION
REFLECTED RADIATION (ALBEDO)
PHOTOVOLTAIC PANEL
REFLECTED RADIATION (ALBEDO)
SEA
EARTH

Solar energy

- **Solar energy** is radiant light and heat from the Sun that is harnessed using a range of ever-evolving technologies(electro magnetic radiation).
- It is an important source of renewable energy and its technologies are broadly characterized as either passive solar or active solar depending on how they capture and distribute solar energy or convert it into solar power.
- Energy used for cooling/heating/drying/distillation/power generation ,etc.,
- The Earth receives 174,000 terawatts (TW) of incoming solar radiation (insolation) at the upper atmosphere.
- Most of the world's population live in areas with insolation levels of 150-300 watts/m^2, or 3.5-7.0 kWh/m^2 per day.

Solar Radiations

- The energy radiated from the sun is electromagnetic waves reaching the planet earth in three spectral regions;
- (i) Ultraviolet 6.4 % ($\lambda \leq 0.38$ µm),
- (ii) Visible 48.0 % (0.38 µm $< \lambda < 0.78$ µm), and
- (iii) Infrared 45.6 % ($\lambda \geq 0.78$ µm) of total energy.
- Due to the large distance between the sun and the earth (1.495 × 108 km) the beam radiation received from the sun on the earth is almost parallel.

- It is important to measure solar radiation:-
- Owing to increasing the number of solar heating and Cooling application.
- For accurate solar radiation data to predict performance.
- To know the availability of the solar energy.

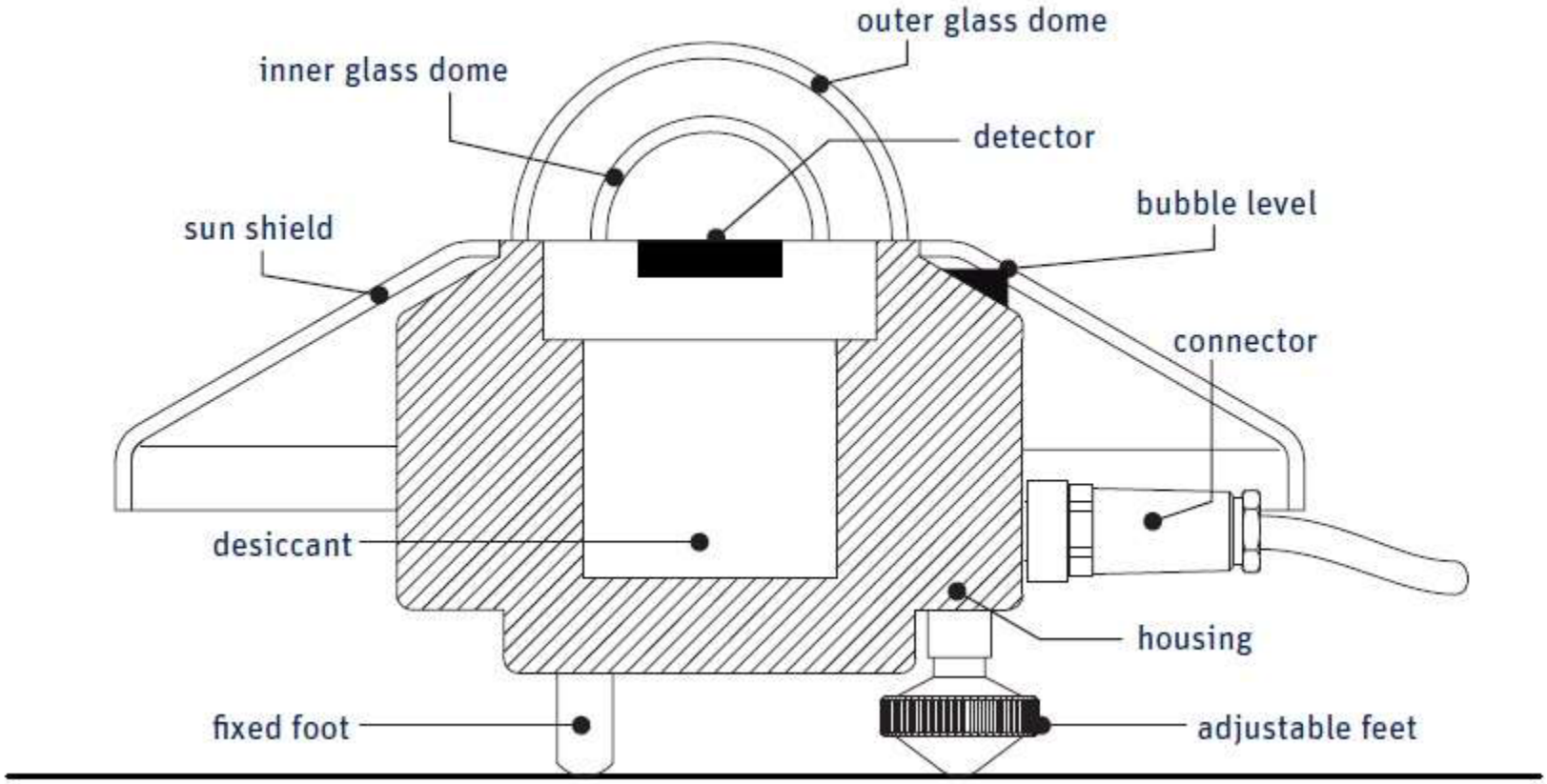
Different Radiation

- **Direct Radiation:** Solar radiation that reaches to the surface of earth without being diffused is called direct beam radiation. (Ib)
- **Diffused Radiation:** As sunlight passes through the atmosphere, some of it is absorbed, scattered and reflected by air molecules, water vapour, cloud, dust, and pollutants from power plants, forest fires, and volcanoes. This is called diffused radiation. (Id)
- **Global Solar Radiation:** The sum of diffuse and direct solar radiation is called global solar radiation. (Ig)

- **ENVIRONMENTAL IMPACT OF SOLAR POWER**
- Every type of energy utilization for electricity generation has environmental consequences, including renewable energies.
- Such as follows:
- manufacture processes
- aesthetic impact
- use of large areas of land
- impact on the eco-system

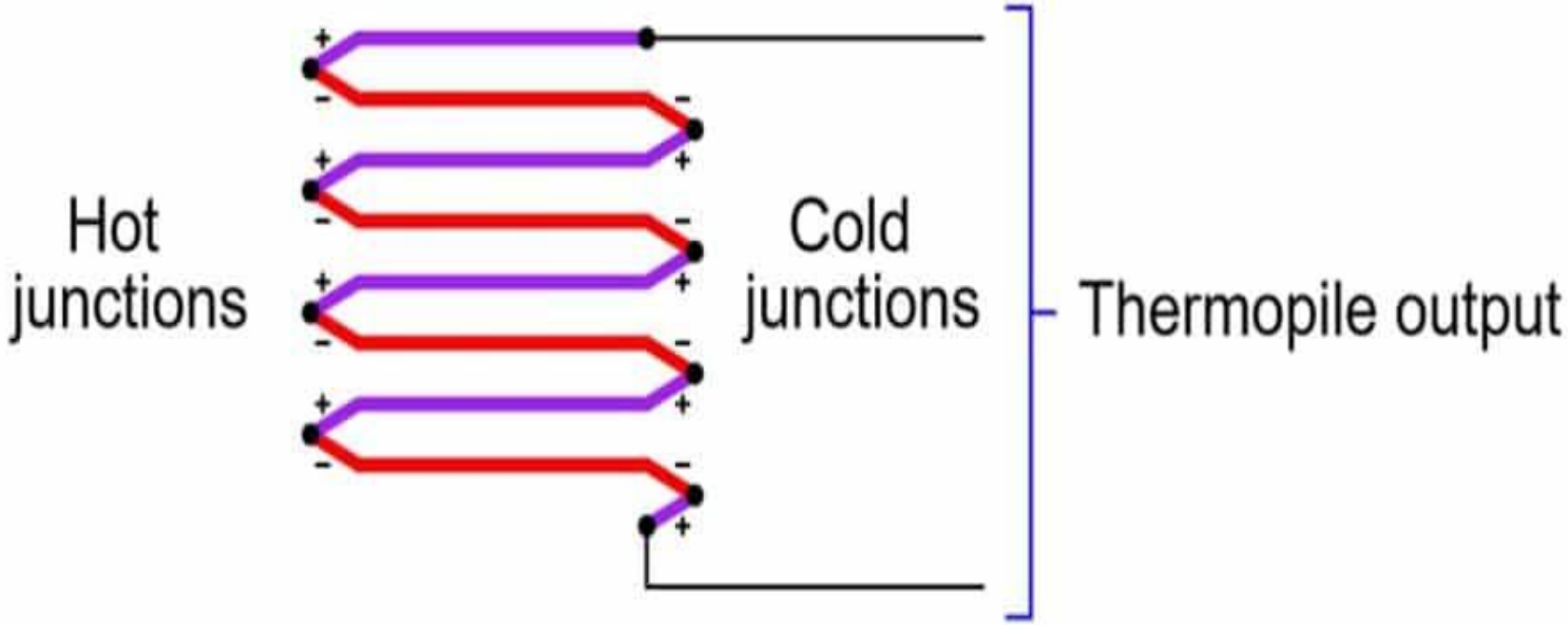
PYRANOMETER

- It measure the total hemispherical solar radiation.
- Solar constant-The average amount of solar radiation received by the Earth's atmosphere, per unit area, when the Earth is at its mean distance from the Sun.
- The solar constant is approximately 1370W/m2.
- The World Metrological Organization (WMO) promotes a value of 1367W/m2
- The NASA design standard for the solar constant is approximately 1353$W/m2$

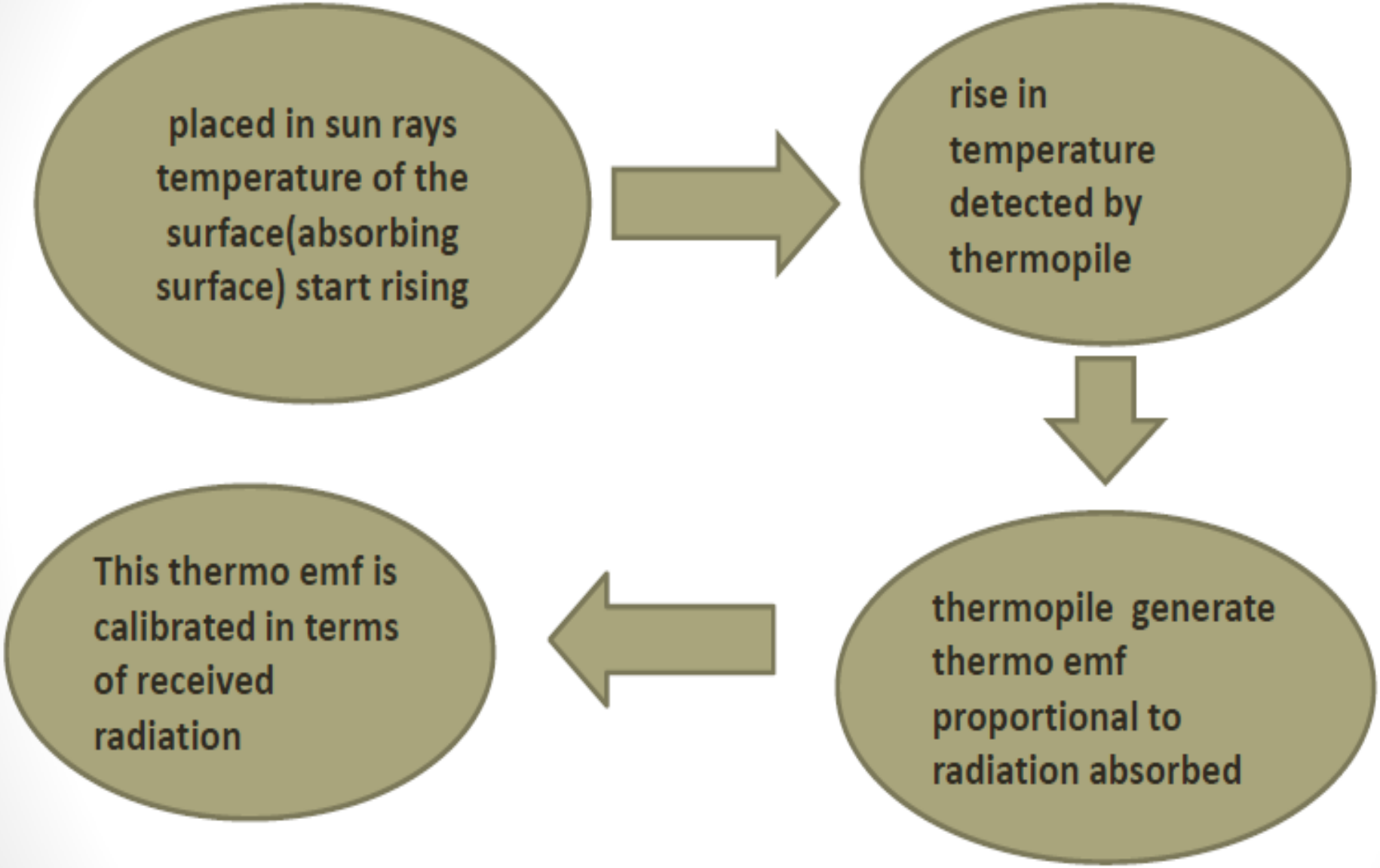

outer glass dome
inner glass dome
detector
bubble level
sun shield
connector
desiccant
housing
fixed foot
adjustable feet

- When it placed in sun rays then it receive the radiation, due to which the temperature of the surface(absorbing surface) start rising .
- The rise in temperature of the absorbing surface is detected by thermopile.
- Now the thermopile generate thermo emf proportional to radiation absorbed.
- This thermo emf is calibrated in terms of received radiation

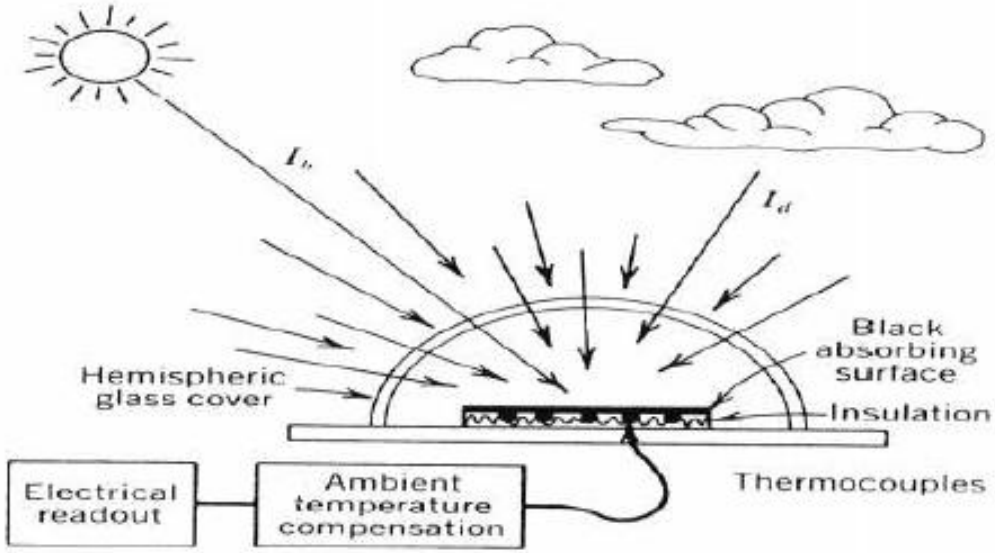
Thermopile

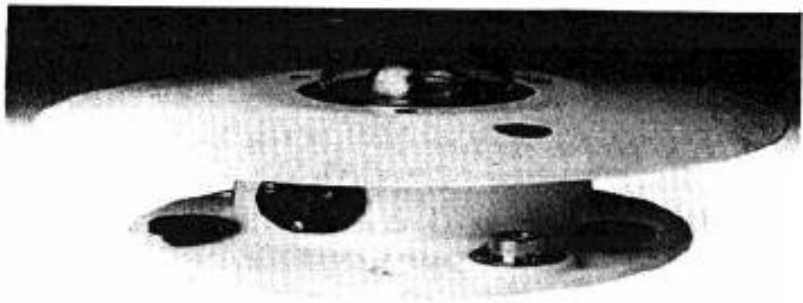


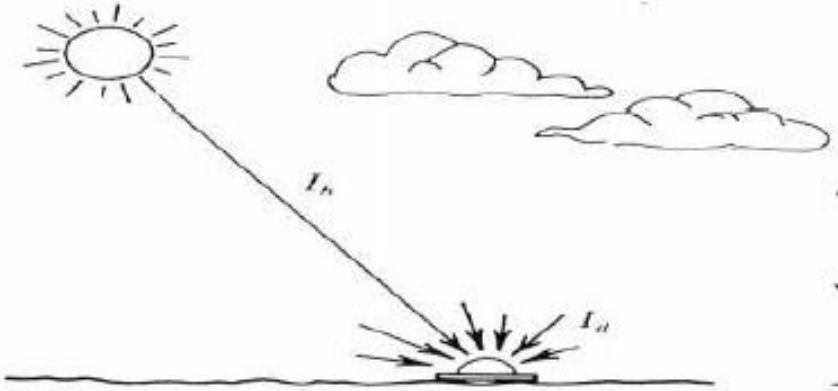
Structure, working principle & Application



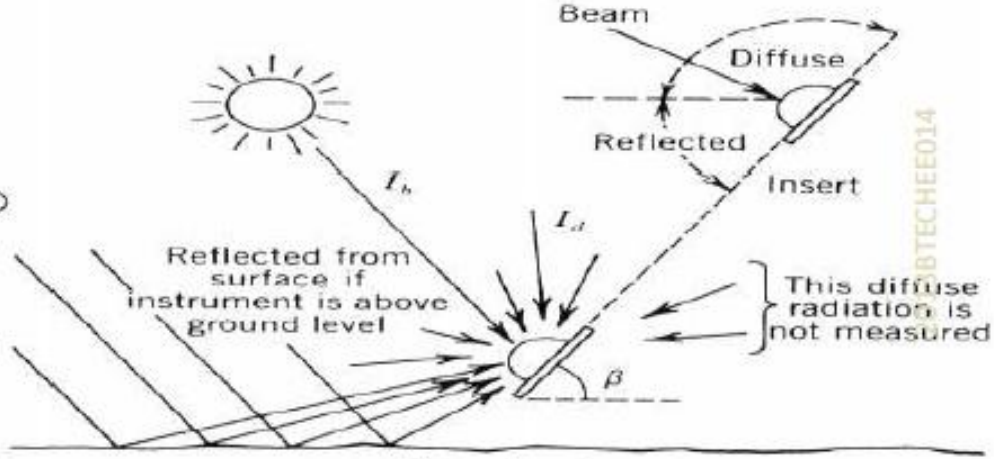
- **Principle**-working principle is that sensitive surface is exposed to total radiation.



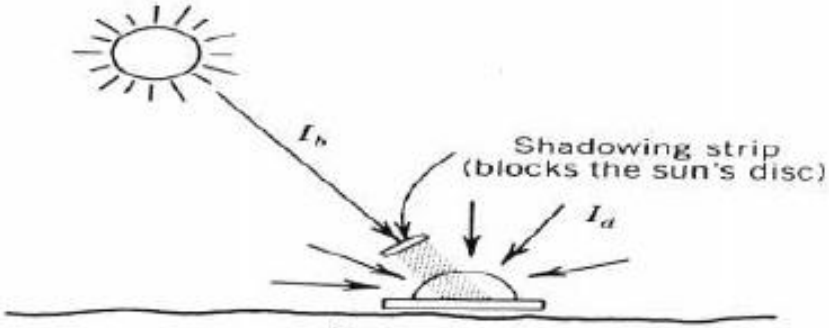




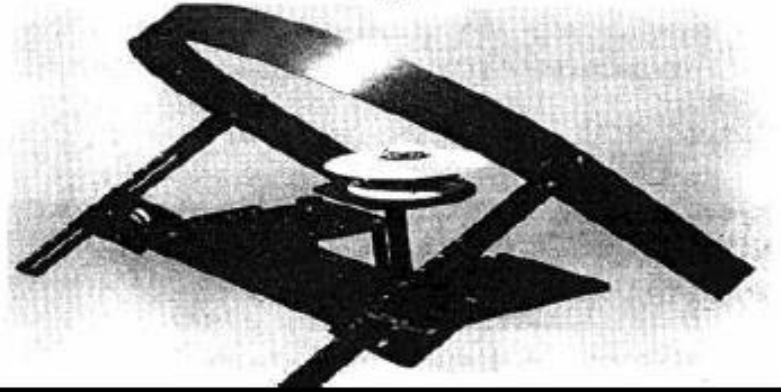
Measuring $I_{t,b}$



Measuring $I_{t,d}$



Measuring $I_{t,d}$



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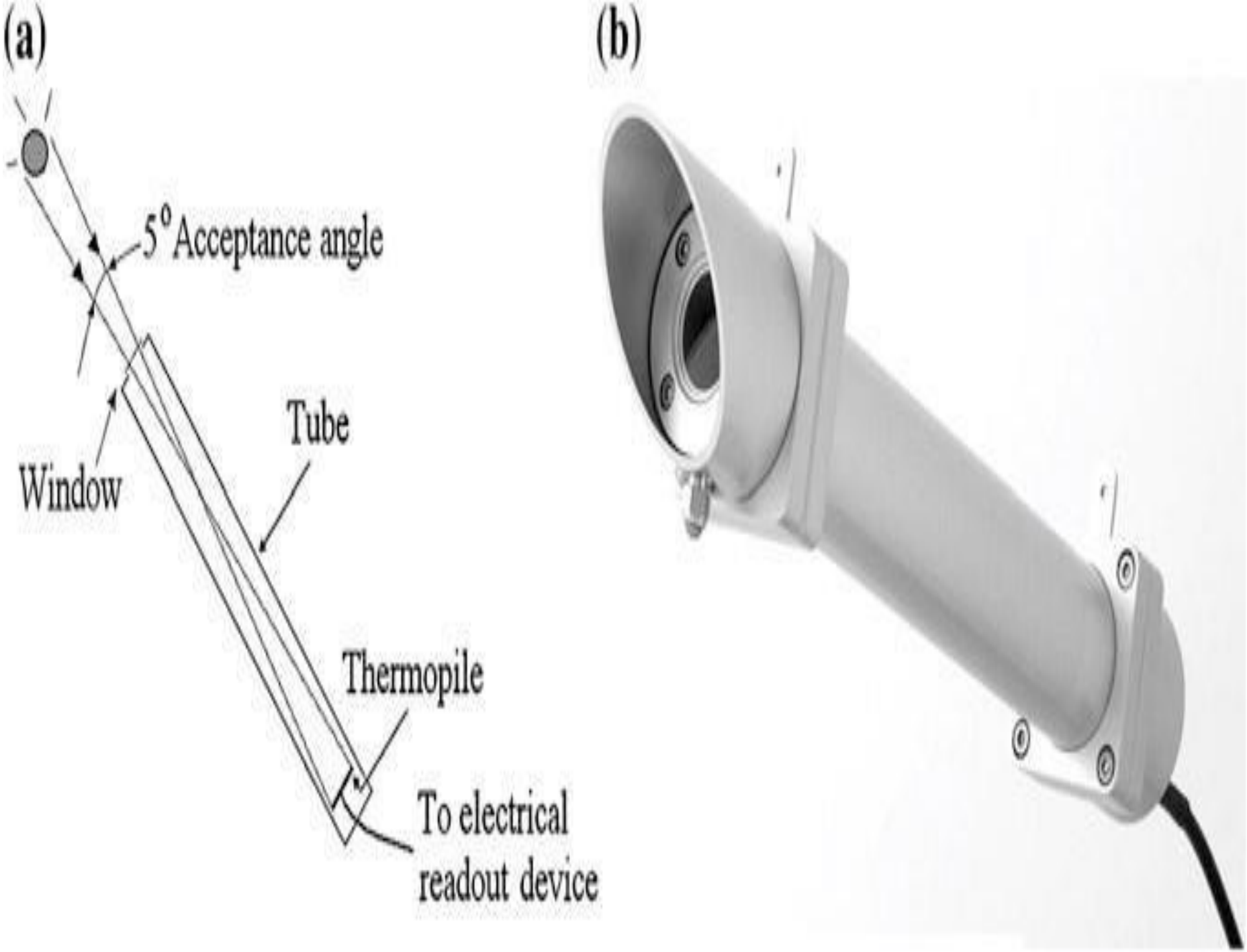
PYROHELIOMETER

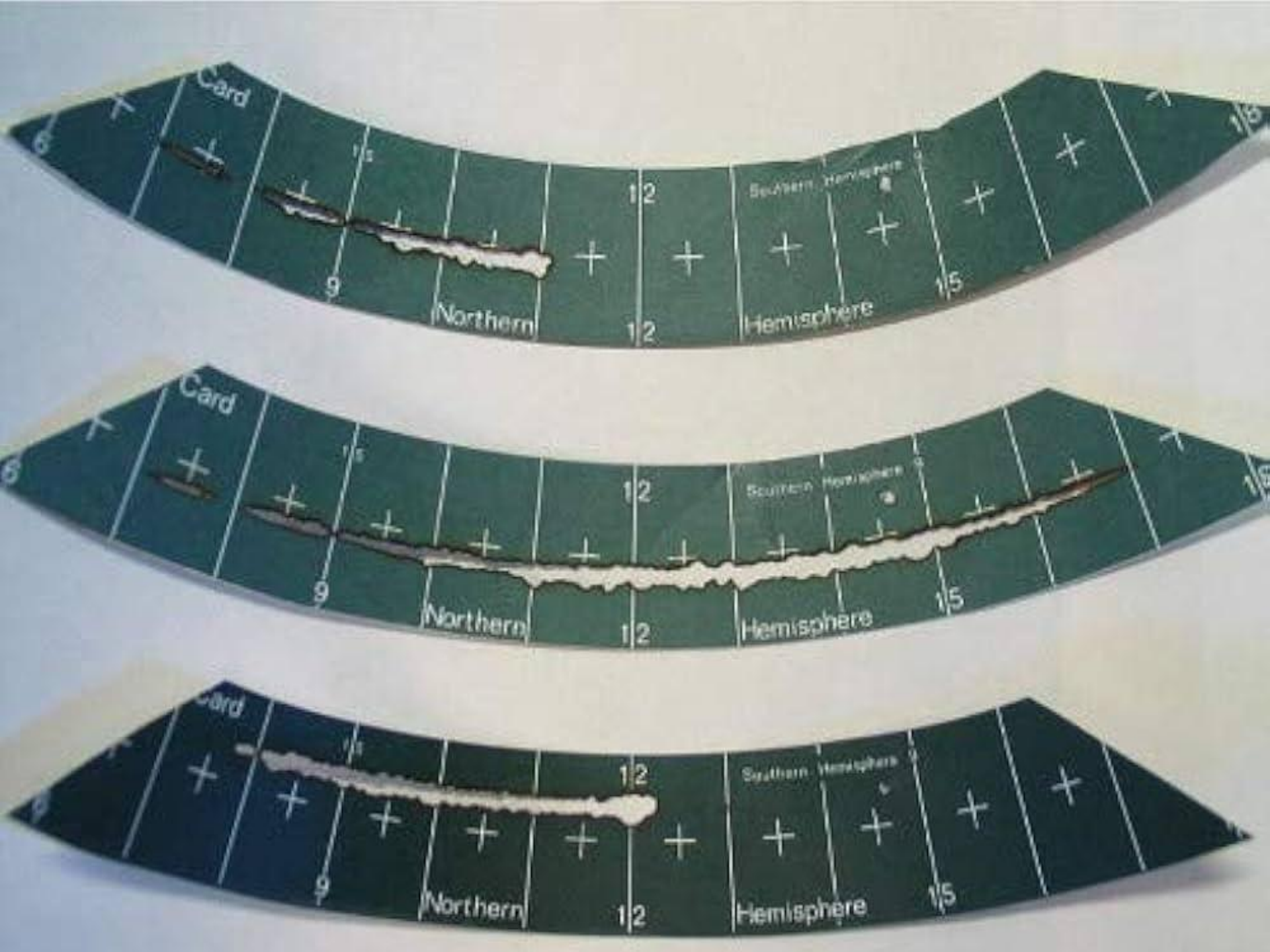
- Used to measure beam of direct radiation.
- It calculate the radiation to determine the beam intensity as a function of incident angle.
- Also known as normal incidence pyranometer.
- Sunlight enters the instrument through collimator tube and is directed onto a thermopile (sensing element) which converts heat to an electrical signal that can be recorded.

- Uses a long collimator tube to collect beam radiation whose field of view is limited to a solid angle of 5.5° by appropriate diaphragms inside the tube.
- Inner surface is blackened to absorb any radiation.
- At the base of tube a wire wound thermopile having a sensitivity of approximately 8 micro volt/W/M^2
- Tube sealed with dry air to eliminate absorption of beam of radiation within the tube by water vapour.


(a)
5° Acceptance angle
Tube
Window
Thermopile
To electrical
readout device
(b)

Sunshine recorder

- Measures the duration in hours of bright sunshine during the course of the day. Consist of a glass sphere installed in a section of “spherical metal bowl” having groves for holding a recorder card strip and the glass sphere.


Card
Northern
Hemisphere
Card
Northern
Hemisphere
Card
Northern
Hemisphere

- Small ‘A’ Type Curved Card is used from January 1 to February 28 or 29 after this period again this card is used from October 15 to December 31. Thus, a total of 137 to 138 cards ‘A’ are used.
- Straight ‘B’ Type Cards are used from March 1 to April 11 and again from September 3 to October 14. A total of 84 Cards is used.
- Long Curved ‘C’ Type Cards are used from April 12 to September 2 a total period of 144 days.